

Question 1 — Science

A simple electric circuit is a closed path in which the electric current can flow from one point to another. The main components used in this circuit are a battery, a switch, a bulb, a resistance and measuring instruments like ammeter and voltmeter.

Q1.C3

The battery is the source of energy and it provides the potential difference which pushes the current in the circuit. The switch is used for making the circuit open or close. When the switch is closed the path becomes complete and the current flows through the bulb and the resistor and the bulb glows. When the switch is open the path is broken from that point so no current can flow and the bulb does not glow.

Q1.C1

In my diagram the battery, switch, resistor, bulb and the ammeter are all joined one after another in series. The ammeter is connected in series because it has to measure the current which is passing through the circuit.

Q1.C2

The voltmeter is also connected in series in the same loop so that it can measure the voltage of the circuit.

The voltmeter should be connected in parallel across the bulb, not in series in the main loop.

Q1.C5

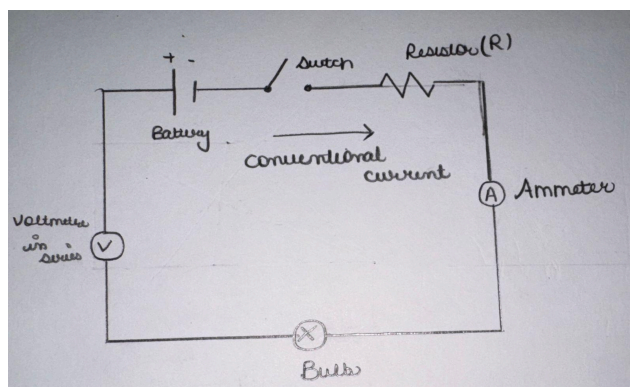
The current flows from the positive terminal of the battery, through the switch, resistor, ammeter and bulb, and returns back to the negative terminal. This is called the conventional current direction and I have shown it by an arrow in my diagram.

Ensure that the physical laws and component placements described in your explanation are scientifically accurate to maintain logical coherence.

Q1.C4

The current in the circuit depends on the voltage and the resistance. If we increase the resistance in the circuit then more current will flow through the bulb, because the resistance pushes the current forward with more force. So a higher resistance will give a brighter bulb.

According to Ohm's law, an increase in resistance results in a decrease in current ($I = V/R$), assuming voltage remains constant.



Question 2 — English

Q2.C1

In my view technology has made students more dependent on easy answers rather than making them better learners.

Q2.C2

Earlier a student had to sit with a problem and think about it for a long time, but now the answer is available in few seconds. Because of this the habit of struggling with a question is slowly finishing.

Q2.C4

For example, in my own class many students directly search the solution of maths sums online and copy the steps without understanding why those steps are used. When the exam comes they are not able to solve a similar sum on their own.

Q2.C3

It is true that technology also give many benefits. A student who does not understand a topic in the class can watch a video at home and understand it better, and digital libraries has made information available to those students who cannot buy costly books. This is a real advantage and I am not denying it.

Q2.C5

But overall I feel technology is making learning much better for students today and every student should use it as much as possible.

Ensure your conclusion aligns with and summarizes the arguments you developed throughout your essay, rather than introducing a contradictory stance at the end.

Question 3 — Economics

Q3.C1

(a) The demand and supply graph is drawn below. Price is taken on the vertical axis and quantity is taken on the horizontal axis. The demand curve is going downward from left to right because when the price is less the consumers buy more quantity. The supply curve is going upward from left to right because when the price is high the producers are ready to supply more.

Q3.C3

(c) When the market price is below the equilibrium price, the quantity supplied becomes more than the quantity demanded and this creates a surplus in the market. When the market price is above the equilibrium price, the quantity demanded becomes more than the quantity supplied and this creates a shortage in the market.

Below equilibrium price, there is a shortage ($QD > QS$). Above equilibrium price, there is a surplus ($QS > QD$).

Q3.C2

(b) The equilibrium point is the place where the demand and supply curves intersect. From the table and from my graph the equilibrium of the market comes at the price of ₹40 and the quantity of 40 units, because this is the point where the two curves are meeting each other. *written below by mistake

The equilibrium occurs at a price of Rs.30 and a quantity of 60 units, where the demand and supply curves intersect.

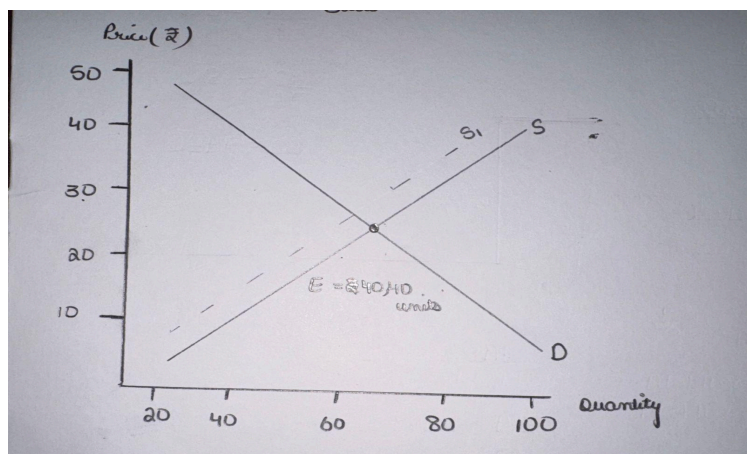
Q3.C4

(d) If the cost of production of the product increases then the producers will not find it profitable to produce the same quantity at the old price. So they will supply less at every price and the supply curve will shift towards the left side. I have shown this new supply curve as S_1 in my graph.

Q3.C5

After this shift the new equilibrium will come at a lower price and a lower quantity than before.

The new equilibrium should result in a higher equilibrium price and a lower equilibrium quantity.



Confidence 0.98 · No review required

Please pay close attention to spelling and grammar errors, such as 'resistance' (resistance), 'amether' (ammeter), 'voltmetre' (voltmeter), and 'equilibrium' (equilibrium). In Science, review Ohm's law and the proper placement of measuring instruments. In English, ensure your conclusion is logically consistent with your thesis. In Economics, review how market forces resolve shortages and surpluses, and how supply shifts affect equilibrium price.

Criteria

Q1.C1 1 / 1 CORRECT confidence 0.95

Correctly represents the main circuit with battery, switch, bulb and resistor connected in a closed series path

"In my diagram the battery, switch, resistor, bulb and the amether are all joined one after another in series."

You have correctly represented the main circuit components (battery, switch, resistor, and bulb) in a closed series path.

Q1.C2 0 / 1 INCORRECT confidence 1.00

Correct placement of ammeter in series and voltmeter in parallel across the bulb

"The voltmetre is also connected in series in the same loop so that it can measure the voltage of the circuit."

While you correctly placed the ammeter in series, you incorrectly placed the voltmeter in series as well. A voltmeter must be connected in parallel across the component (the bulb) to measure its potential difference.

Correction: The voltmeter should be connected in parallel across the bulb, not in series in the main loop.

Q1.C3 1 / 1 CORRECT confidence 0.95

Correct explanation of current flow and the function of the main components

"The battery is the source of energy and it provides the potential difference which pushes the current in the circuit. The switch is used for making the circuit open or close."

You provided a clear and correct explanation of how current flows and the functions of the battery and switch in opening and closing the circuit.

Q1.C4 0 / 1 INCORRECT confidence 1.00

Correctly explains the relationship between resistance and current, including the relevant principle/Ohm's law

"If we increase the resistance in the circuit then more current will flow through the bulb, because the resistance pushes the current forward with more force."

Your explanation of the relationship between resistance and current is incorrect. According to Ohm's law ($V = IR$), if voltage remains constant, increasing the resistance will decrease the current flowing through the circuit, not increase it.

Correction: According to Ohm's law, an increase in resistance results in a decrease in current ($I = V/R$), assuming voltage remains constant.

Q1.C5 0 / 1 PARTIAL confidence 0.90

Clear, logically structured explanation with appropriate labels and current direction in the diagram

"The current flows from the positive terminal of the battery, through the switch, resistor, ammeter and bulb, and returns back to the negative terminal. This is called the conventional current direction and I have shown it by an arrow in my diagram."

Your diagram includes appropriate labels and correctly shows the conventional current direction. However, the overall explanation is not logically sound due to the incorrect placement of the voltmeter in series and the incorrect explanation of the relationship between resistance and current.

Correction: Ensure that the physical laws and component placements described in your explanation are scientifically accurate to maintain logical coherence.

Q2.C1 1 / 1 CORRECT confidence 1.00

Presents a clear position on whether/how technology affects learning

"In my view technology has made students more dependent on easy answers rather than making them better learners."

You have presented a very clear and direct position on how technology affects learning.

Q2.C2 1 / 1 CORRECT confidence 0.95

Provides relevant and logically developed arguments supporting the position

"Earlier a student had to sit with a problem and think about it for a long time, but now the answer is available in few seconds. Because of this the habit of struggling with a question is slowly finishing."

You provided a logically developed argument about how instant access to answers reduces the habit of critical thinking and problem-solving.

Q2.C3 1 / 1 CORRECT confidence 1.00

Recognises and meaningfully addresses an opposing viewpoint or limitation

"It is true that technology also give many benefits. A student who does not understand a topic in the class can watch a video at home and understand it better, and digital libraries has made information available to those students who cannot buy costly books."

You successfully recognized and addressed the opposing viewpoint by discussing the benefits of technology, such as educational videos and digital libraries.

Q2.C4 1 / 1 CORRECT confidence 1.00

Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims

"For example, in my own class many students directly search the solution of maths sums online and copy the steps without understanding why those steps are used."

You used a highly relevant, real-world example from your own classroom experience to support your argument about dependency on online solutions.

Q2.C5 0 / 1 INCORRECT confidence 0.95

Provides a coherent conclusion that follows from the discussion, with clear overall communication

"But overall I feel technology is making learning much better for students today and every student should use it as much as possible."

Your conclusion is not coherent because it directly contradicts your initial position and the main arguments you developed. You argued that technology makes students dependent and worse learners, but concluded that it makes learning much better and should be used as much as possible.

Correction: Ensure your conclusion aligns with and summarizes the arguments you developed throughout your essay, rather than introducing a contradictory stance at the end.

Q3.C1 1 / 1 CORRECT confidence 0.95

Correctly plots and labels the demand and supply curves, with appropriate axes and direction

"The demand and supply graph is drawn below. Price is taken on the vertical axis and quantity is taken on the horizontal axis."

You have correctly plotted and labeled the demand and supply curves, with appropriate axes, labels, and directions.

Q3.C2 0 / 1 INCORRECT confidence 1.00

Correctly identifies the equilibrium at Rs.30 and 60 units and explains why it is equilibrium

"From the table and from my graph the equilibrium of the market comes at the price of Rs.40 and the quantity of 40 units, because this is the point where the two curves are meeting each other."

You incorrectly identified the equilibrium point. Based on the data and your own graph, the curves intersect at a price of Rs.30 and a quantity of 60 units, where quantity demanded equals quantity supplied.

Correction: The equilibrium occurs at a price of Rs.30 and a quantity of 60 units, where the demand and supply curves intersect.

Q3.C3 0 / 1 INCORRECT confidence 1.00

Correctly explains shortage below equilibrium and surplus above equilibrium

"When the market price is below the equilibrium price, the quantity supplied becomes more than the quantity demanded and this creates a surplus in the market. When the market price is above the equilibrium price, the quantity demanded becomes more than the quantity supplied and this creates a shortage in the market."

You have reversed the concepts of shortage and surplus. When the price is below equilibrium, quantity demanded exceeds quantity supplied, creating a shortage. When the price is above equilibrium, quantity supplied exceeds quantity demanded, creating a surplus.

Correction: Below equilibrium price, there is a shortage ($QD > QS$). Above equilibrium price, there is a surplus ($QS > QD$).

Q3.C4 1 / 1 CORRECT confidence 1.00

Correctly explains that increased production costs shift the supply curve left/upward

"If the cost of production of the product increases then the producers will not find it profitable to produce the same quantity at the old price. So they will supply less at every price and the supply curve will shift towards the left side."

You correctly explained that an increase in production costs reduces profitability, causing the supply curve to shift to the left.

Q3.C5 0 / 1 INCORRECT confidence 1.00

Correctly explains the resulting tendency toward a higher equilibrium price and lower equilibrium quantity, with the change represented appropriately on the graph

"After this shift the new equilibrium will come at a lower price and a lower quantity than before."

Your explanation of the new equilibrium is incorrect. A leftward shift of the supply curve (with demand remaining constant) leads to a higher equilibrium price and a lower equilibrium quantity, not a lower price.

Correction: The new equilibrium should result in a higher equilibrium price and a lower equilibrium quantity.